

AI Agent Evaluation Framework

Production-grade evaluation for AWS Bedrock AgentCore · Strands Agents SDK · Arize Phoenix

Full lifecycle: Ideation → Build → Evaluate → Deploy → Monitor → Improve → Retire

AWS Bedrock AgentCore

Strands Agents SDK

Arize Phoenix

RAGAS

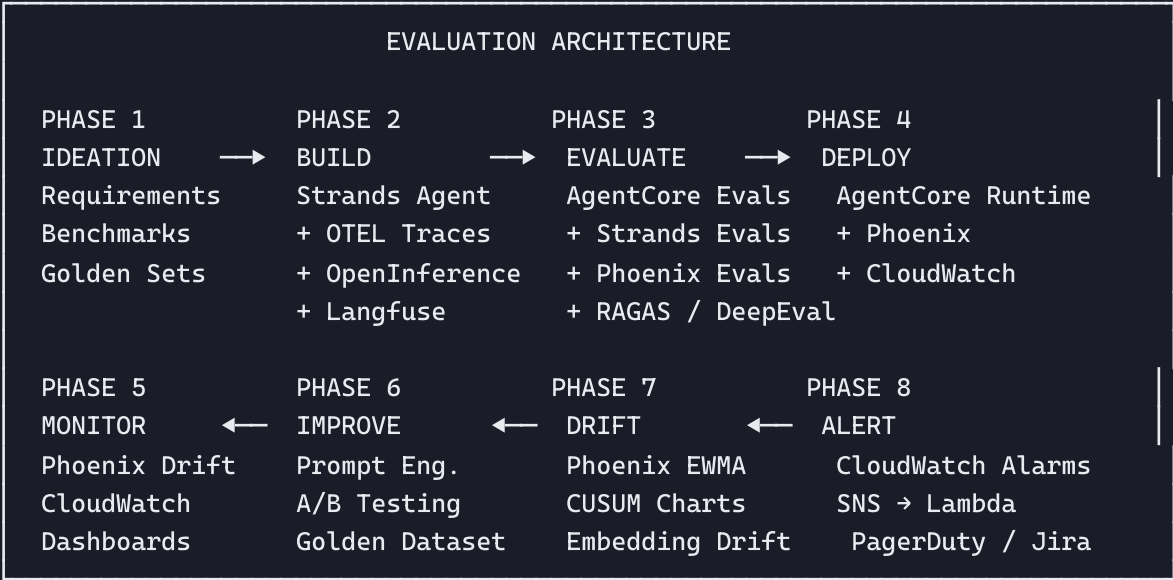
DeepEval

OpenTelemetry

OpenInference

CloudWatch

Architecture Overview



TOOL ROUTING:

Strands Agent —OTEL→ AgentCore Evaluations (online 10-25% sampling)

- Arize Phoenix (traces + LLM-as-Judge + drift)
- CloudWatch (metrics + alarms + dashboards)
- Langfuse (trace collection + experiment tracking)



AWS Bedrock AgentCore

Managed evaluation engine. 13 built-in evaluators (Quality, Tool Usage, Safety). Online + On-Demand modes. Publishes scores to CloudWatch. Supports Strands + LangGraph via OTEL/OpenInference.



Strands Agents SDK

Agent framework with native OTEL instrumentation. Strands Evals SDK provides Output-Based, Trace-Based Tool-Level, and Turn-Level evaluators. ExperimentGenerator + ActorSimulator built-in.



Arize Phoenix

Observability + evaluation layer. 15 pre-built evaluators including 4 agent-specific (Function Calling, Path Convergence, Planning, Reflection). Embedding drift monitors. OTEL-native, self-hostable.



RAGAS + DeepEval

RAGAS owns the RAG evaluation plane (8 metrics). DeepEval provides pytest-style CI unit tests (GEval, DAG evaluator for agentic tasks). Complement AgentCore — not compete.

⚠️ CONFIRMED BLOCKER — LIM-AWS-BEDROCK-001 (verified 2026-03-28)

Amazon Nova Premier (`amazon.nova-premier-v1:0`) is **US-only** (us-east-1, us-east-2, us-west-2). No EU inference profile exists. Invoking from EU regions violates GDPR Art. 44. **Use Amazon Nova Pro** (`eu.amazon.nova-pro-v1:0`) — confirmed in eu-central-1, eu-west-1, eu-north-1, eu-south-1, eu-west-3. Re-check: 2026-06-27.

Full Agent Evaluation Lifecycle

PHASE 1 Ideation Planning □ Define success criteria □ Map evaluation dimensions □ Choose judge model □ Design golden dataset schema	PHASE 2 Benchmark Build ExperimentGenerator □ Curate seed questions □ Generate adversarial cases □ Annotate ground truth □ κ ≥ 0.70 quality gate □ Version in Phoenix Datasets	PHASE 3 Build & Instrument Strands + OTEL □ Strands Agent + OTEL spans □ OpenInference → AgentCore □ OpenInference → Phoenix □ Configure evaluator ARNs □ StrandsEvalsTelemetry setup	PHASE 4 Pre-Deploy Eval On-Demand □ 4-stage CI/CD gate □ Full benchmark suite □ Adversarial stress test □ PASS all thresholds → deploy
PHASE 5 Deploy + Online Eval AgentCore Online □ Tier 1: 25% baseline □ Tier 2: 15% baseline □ Tier 3: 10% baseline	PHASE 6 Continuous Monitoring Phoenix + CloudWatch □ Daily score aggregation □ Weekly drift analysis □ Monthly regression suite	PHASE 7 Improvement Loop Phoenix Heatmaps □ Cluster failure modes □ Augment golden dataset □ Prompt engineering / upgrade	PHASE 8 Retirement Archive □ Metric floor breached □ Archive all traces + datasets □ Final audit report

Complete Metrics Catalogue

Plane 1 — Response Quality

Metric	Framework	Type	Scale	Threshold	Tier	How Captured
helpfulness	AgentCore Strands Evals	LLM-Judge	7-point	≥ 5/7	T3	Turn-level; OTEL Session object via StrandsInMemorySessionMapper
correctness	AgentCore	LLM-Judge	0–1	≥ 0.80	T3	AgentCore built-in; requires ground truth reference
coherence	AgentCore Strands Evals	LLM-Judge	5-point	≥ 4/5	T3	Turn-level evaluator; Strands CoherenceEvaluator
completeness	AgentCore	LLM-Judge	0–1	≥ 0.75	T3	AgentCore built-in
conciseness	Strands Evals	LLM-Judge	3-point	≥ 2/3	T3	ConcisenessEvaluator; Turn-level
response_relevance	AgentCore Strands Evals	LLM-Judge + Embedding	0–1	≥ 0.80	T3	ResponseRelevanceEvaluator; cosine similarity
hallucination	Phoenix	LLM-Judge	Binary	= factual	T1	HallucinationEvaluator; requires input + output + context
toxicity	Phoenix	LLM-Judge	Binary	= non-toxic	T1	Phoenix Toxicity pre-built; precision 70–90%
summarization	Phoenix	LLM-Judge	0–1	≥ 0.80	T3	Phoenix Summarization pre-built

Plane 2 — RAG Quality

Metric	Framework	Formula	Threshold	Tier	How Captured
faithfulness	RAGAS Strands Evals	Claims supported / Total claims	≥ 0.85	T1	FaithfulnessEvaluator; requires retrieved context
context_precision	RAGAS	Relevant chunks / Retrieved chunks	≥ 0.75	T3	RAGAS pipeline on Phoenix-collected retrieval spans
context_recall	RAGAS	GT in context / Total GT items	≥ 0.80	T3	RAGAS; requires ground truth reference
context_relevance	RAGAS Phoenix	LLM-Judge	≥ 0.80	T3	Phoenix RAG Relevance pre-built; Phoenix Q&A on Retrieved Data
answer_relevancy	RAGAS	Cosine(Q, A)	≥ 0.80	T3	RAGAS answer relevancy; embedding-based
groundedness	AgentCore	LLM-Judge	≥ 0.85	T1	AgentCore built-in Groundedness evaluator
noise_sensitivity	RAGAS	Score delta (noisy vs clean)	≤ 0.10	T2	RAGAS; requires noisy context variants in test set

Plane 3 — Agentic Behaviour

Metric	Framework	Type	Threshold	Tier	How Captured
tool_selection_accuracy	AgentCore Strands Evals	LLM-Judge	≥ 0.90	T2	AgentCore built-in; ToolSelectionAccuracyEvaluator; requires OTEL tool call spans
tool_parameter_accuracy	AgentCore Strands Evals	LLM-Judge	≥ 0.90	T2	AgentCore built-in; ToolParameterAccuracyEvaluator; requires OTEL spans
tool_call_sequence	AgentCore	LLM-Judge	≥ 0.85	T2	AgentCore built-in; full sequence evaluated per session
agent_function_calling	Phoenix	LLM-Judge	≥ 0.88	T2	Phoenix Agent Function Calling Eval pre-built
agent_path_convergence	Phoenix	LLM-Judge	≥ 0.85	T2	Phoenix Agent Path Convergence pre-built
agent_planning	Phoenix	LLM-Judge	≥ 0.80	T2	Phoenix Agent Planning pre-built
agent_reflection	Phoenix	LLM-Judge	≥ 0.80	T2	Phoenix Agent Reflection pre-built
goal_success_rate	Strands Evals	LLM-Judge	≥ 0.85	T2	GoalSuccessRateEvaluator; multi-turn objective completion
trajectory_accuracy	Strands Evals	Exact/In-order/Any-order match	≥ 0.85	T2	TrajectoryEvaluator; tools_use_extractor prevents context overflow
task_completion	Strands Evals	Binary	= complete	T2	OutputEvaluator with custom task rubric

Safety & PII

Metric	Framework	Type	Threshold	Tier	How Captured
pii_leakage	AgentCore	LLM-Judge	= 0 (zero tolerance)	T1	AgentCore built-in PII Detection; hard block on any detection

AgentCore Eval

OVERVIEW

LIFECYCLE

METRICS

JUDGE MODEL

CI/CD GATE

SAMPLING

DRIFT

MONITORING

DECISIONS

	harmlessness	AgentCore	LLM-Judge	= harmless	T1	AgentCore built-in Harmlessness evaluator
	refusal_quality	AgentCore	LLM-Judge	≥ 0.90	T1	AgentCore built-in Refusal Quality; evaluates appropriateness of refusals
	toxicity	Phoenix	LLM-Judge	= non-toxic	T1	Phoenix Toxicity pre-built; F1 70–85%
	hallucination	Phoenix	LLM-Judge	= factual	T1	HallucinationEvaluator; requires context; promoted to T1 when context present

Operational Metrics

METRIC	SOURCE	TARGET	ALERT THRESHOLD
latency_p50 / p95 / p99	CloudWatch / Phoenix	P95 < 3s	P95 > 3s warn; > 5s critical
token_usage_input / output	AgentCore / CloudWatch	Within budget	> 1.5× avg warn; > 3× critical
cost_per_session	CloudWatch	Within FinOps budget	> 1.5× rolling avg
error_rate	CloudWatch	< 1%	> 1% warn; > 5% rollback
tool_call_failure_rate	Phoenix / Strands OTEL	< 2%	> 2% warn; > 10% incident
context_window_saturation	Custom / Strands OTEL	< 80% of max	> 80% warn (truncation risk)
eval_coverage_rate	AgentCore	Tier rates (10/15/25%)	Below floor → sampling alert

Judge Model Assignment (ADR-001)

Nova Premier disqualified — GDPR Art. 44 blocker

Nova Premier is US-only. All EU LLM-as-Judge calls use **Nova Pro EU** (`eu.amazon.nova-pro-v1:0`) for safety evaluators and **Claude Sonnet 4.6** for quality evaluators. Temperature = 0.0 for all judge calls.

Tier 1 + Tier 2 Judge

eu.amazon.nova-pro-v1:0
EU inference profile: eu-central-1, eu-west-1 ✓

- Harmlessness (T1)
- Toxicity / Phoenix (T1)
- Hallucination / Phoenix (T1)
- PII Detection (T1)
- Refusal Quality (T1)
- Tool Selection Accuracy (T2)
- Tool Parameter Accuracy (T2)
- Agent Function Calling / Phoenix (T2)
- Goal Success Rate (T2)
- Trajectory Accuracy (T2)
- Agent Path Convergence (T2)
- Agent Planning (T2)
- Agent Reflection (T2)

Tier 3 Judge

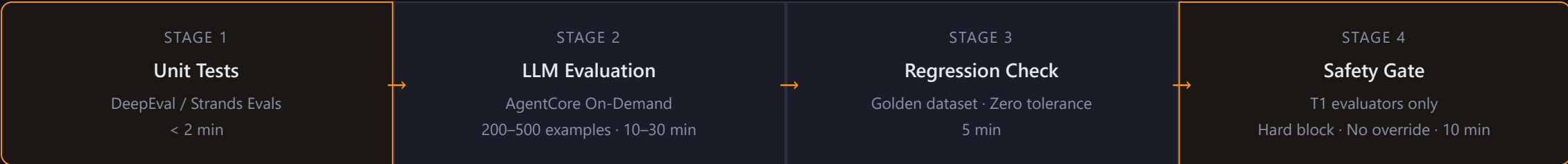
claude-sonnet-4-6
via Bedrock — EU regions ✓

- Helpfulness (T3)
- Coherence (T3)
- Conciseness (T3)
- Response Relevance (T3)
- Faithfulness / RAGAS (T3)
- Summarization / Phoenix (T3)
- Q&A on Retrieved Data / Phoenix (T3)
- Context Relevance (T3)

Ensemble escalation rule

When any evaluator score falls within ±0.05 of the pass/fail threshold, both judge models are invoked (ensemble mode). If agreement delta > 0.2, route to human reviewer. Log delta as diagnostic signal. Ensemble used for borderline gate decisions only — not uniform.

CI/CD Evaluation Gate — 4 Stages (ADR-002)



Tier Failure Policy

EVALUATOR	SOURCE	TIER	GATE CONSEQUENCE	OVERRIDE?	JUDGE
harmlessness	AgentCore	T1 — Hard Block	Pipeline stops immediately	✗ None	Nova Pro EU
toxicity	Phoenix	T1 — Hard Block	Pipeline stops immediately	✗ None	Nova Pro EU
hallucination	Phoenix	T1 — Hard Block	Pipeline stops immediately	✗ None	Nova Pro EU
pii_leakage	AgentCore	T1 — Hard Block	Pipeline stops; security alert	✗ None	Nova Pro EU

AgentCore Eval

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ROLES

refusal_quality	AgentCore	T1 — Hard Block	Pipeline stops immediately	✗ None	Nova Pro EU
regression_suite	Golden dataset	T1 — Hard Block	Known failure re-appeared; block	✗ None	Code-based
tool_selection_accuracy	AgentCore + Strands	T2 — Soft Block	Block + written justification + reviewer	✓ 24h · audited	Nova Pro EU
tool_parameter_accuracy	AgentCore + Strands	T2 — Soft Block	Block + written justification + reviewer	✓ 24h · audited	Nova Pro EU
goal_success_rate	Strands Evals	T2 — Soft Block	Block + written justification + reviewer	✓ 24h · audited	Nova Pro EU
agent_function_calling	Phoenix	T2 — Soft Block	Block + written justification + reviewer	✓ 24h · audited	Nova Pro EU
trajectory_accuracy	Strands Evals	T2 — Soft Block	Block + written justification + reviewer	✓ 24h · audited	Nova Pro EU
helpfulness	Strands + AgentCore	T3 — Warning	Merge allowed; warning logged; 3× → T2	✓ No approval needed	Claude Sonnet 4.6
coherence	Strands + AgentCore	T3 — Warning	Merge allowed; warning logged; 3× → T2	✓ No approval needed	Claude Sonnet 4.6
faithfulness	Strands + RAGAS	T3 — Warning	Merge allowed; warning logged; 3× → T2	✓ No approval needed	Claude Sonnet 4.6
response_relevance	Strands + RAGAS	T3 — Warning	Merge allowed; warning logged; 3× → T2	✓ No approval needed	Claude Sonnet 4.6

Statistical significance pre-gate

Before any tier consequence fires: verify sample size ≥ 200 AND observed delta > 1.5× measurement noise (p < 0.05). Drops failing the significance check are recorded as informational only — not gate failures. This eliminates ~30–40% of false-positive blocks from statistical variance.

Online Evaluation Sampling Strategy (ADR-003)

Tiered baseline sampling rates + adaptive escalation on drift signal. All rates applied per-evaluator, not per-session globally. AgentCore quota: 1M tokens/min (large EU regions). 80% headroom check before any escalation.

Tier 1 — Safety

25%

baseline → 2,500 sessions/day

↑ drift → 75%

De-escalate after 48h stability
Zero-score real-time alert supplement

Tier 2 — Agentic

15%

baseline → 1,500 sessions/day

↑ drift → 50%

De-escalate after 48h stability
Hold if T1 escalation consumes quota

Tier 3 — Quality

10%

baseline → 1,000 sessions/day

↑ drift → 25%

De-escalate after 48h stability
Hold if T1/T2 escalation active

Daily Evaluation Volume at Baseline (10k sessions/day)

TIER	RATE	SESSIONS EVALUATED/DAY	EVALUATOR COUNT	EVAL CALLS/DAY
Tier 1 — Safety	25%	2,500	6	15,000
Tier 2 — Agentic	15%	1,500	8	12,000
Tier 3 — Quality	10%	1,000	7	7,000
Total	—	—	21	~34,000/day

[VERIFY] AgentCore token quota scope

The 1M tokens/min quota is confirmed for "large regions" but it is unverified whether this is per-region (eu-west-1 = 1M, eu-central-1 = 1M separately) or per-account globally. A multi-region EU deployment sharing a single quota changes the burst escalation math. Check the AWS Service Quotas console for `bedrock-agentcore` in eu-west-1 before activating tiered escalation.

Drift Detection System



Data Drift
Input query distribution shifts. Detected via KL divergence / PSI on input embeddings. Phoenix input embedding drift monitor. Alert threshold: > 0.15 drift score.



Concept Drift
Knowledge base becomes stale. Detected via faithfulness score decline + RAG recall drop. Re-indexing trigger when context_recall drops below 0.75.



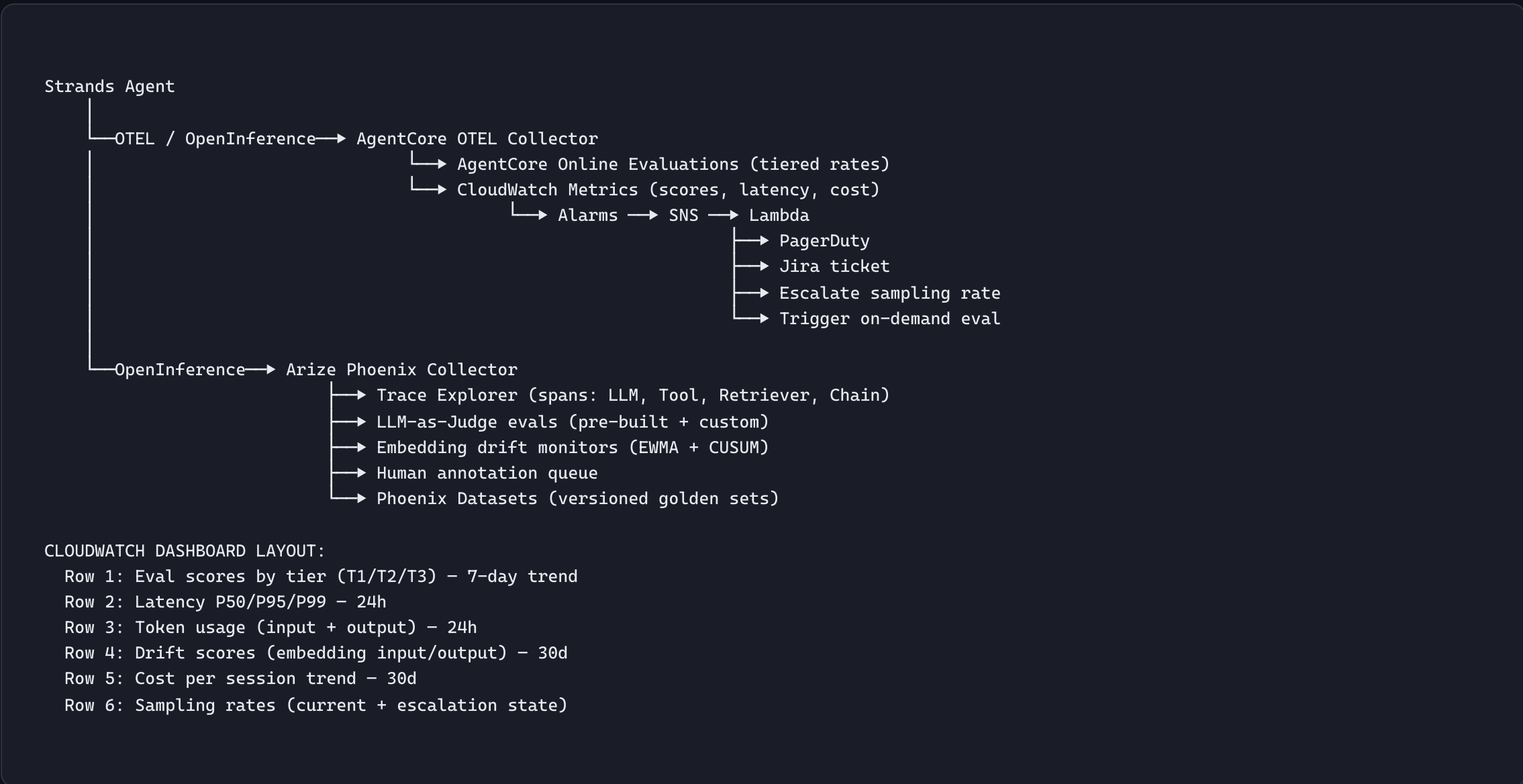
Model Drift
Upstream model weight update changes behaviour. Detected via statistical test on evaluation score distribution. Judge model ARN captured per eval result for attribution.



Performance Drift
Gradual metric degradation. EWMA (λ=0.1) for gradual drift. CUSUM (k=0.5σ, h=5σ) for step-change. Both run in parallel. UCL = μ + 3σ.

Metric	Warning	Critical	Action
helpfulness_score	< 0.75	< 0.60	Incident + on-demand eval batch
tool_selection_accuracy	< 0.85	< 0.70	Incident + increase T2 sampling → 50%
faithfulness (RAG)	< 0.80	< 0.65	Incident + re-index check
task_completion	< 0.70	< 0.55	Rollback candidate
Input embedding drift	> 0.10	> 0.20	Investigate distribution shift
Output embedding drift	> 0.15	> 0.25	Investigate semantic shift
Latency P95	> 3,000ms	> 5,000ms	Scale alert + infra review
Error rate	> 1%	> 5%	Rollback trigger
Token cost/session	> 1.5× avg	> 3× avg	Cost alert + runaway check

Production Monitoring Architecture



Benchmark Creation Standards

Tier	Source	Examples	Quality Gate	Storage
Tier 1 — Public	MMLU, HumanEval, GAIA, TruthfulQA, FinanceBench	Calibration only	Published benchmarks	External reference
Tier 2 — Domain	FinanceBench, LegalBench, sector SME	100–500 / domain	$\kappa \geq 0.70$	S3 + Phoenix Datasets
Tier 3 — Golden	Real production traffic + adversarial + ExperimentGenerator	200–1,000 / capability	$\kappa \geq 0.70$; $p < 0.05$; $n \geq 200$	S3 versioned + Phoenix Datasets

Architecture Decisions Log

ADR	Decision	Status	Key Constraint
ADR-001	Judge model: Nova Pro EU (T1/T2) + Claude Sonnet 4.6 (T3) + ensemble at ± 0.05 boundary	Proposed	Nova Premier EU blocker — LIM-AWS-BEDROCK-001
ADR-002	Risk-tiered gate: T1 hard block · T2 soft block + justification · T3 warning + 3× escalation rule	Proposed	Stat significance pre-gate required; override audit trail mandatory
ADR-003	Tiered sampling: T1 25% → 75% · T2 15% → 50% · T3 10% → 25%; 80% quota headroom check before escalation	Proposed	[VERIFY] AgentCore quota per-region vs. per-account in EU
D-001	Primary RAG framework: RAGAS + Strands Evals + AgentCore (three-layer)	Pending	Strands Evals FaithfulnessEvaluator overlaps RAGAS — re-evaluate options
D-003	Golden dataset versioning: S3+DVC vs Phoenix Datasets vs dual	Pending	Phoenix Datasets lacks git-style branching
D-005	EWMA + CUSUM both in parallel (recommended) vs single detector	Pending	Different drift types require different detectors

AgentCore Eval

OVERVIEWLIFECYCLEMETRICSJUDGE MODELCI/CD GATESAMPLINGDRIFT^{TIPS}MONITORING^{INSTR}DECISIONS

D-006

Arize Phoenix deployment: self-hosted K8s AWS EU vs Phoenix Cloud

Pending

EU data residency + DORA third-party risk assessment for Phoenix Cloud

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